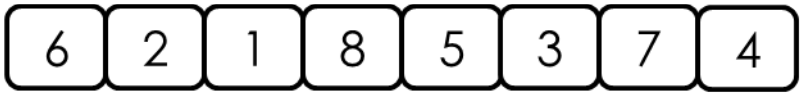
Merge Sort Algorithm

The Merge Sort Algorithm

This sorting algorithm is more efficient than both the 'Bubble' and 'Insertion' sorts. This is because it uses the principles of 'Divide and Conquer'. It works by splitting the dataset up into its individual items and then continually pairing items from the dataset in order until the entire dataset has been ordered.

An Example:

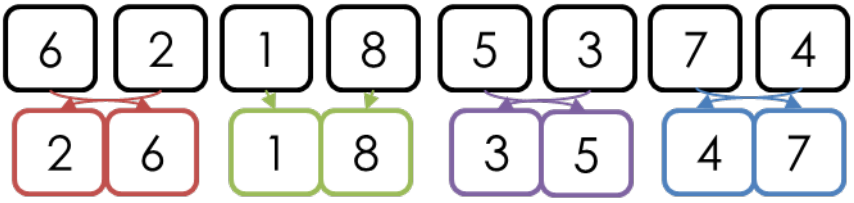
We start with a set of data to be sorted.



The data set is split into its individual items.

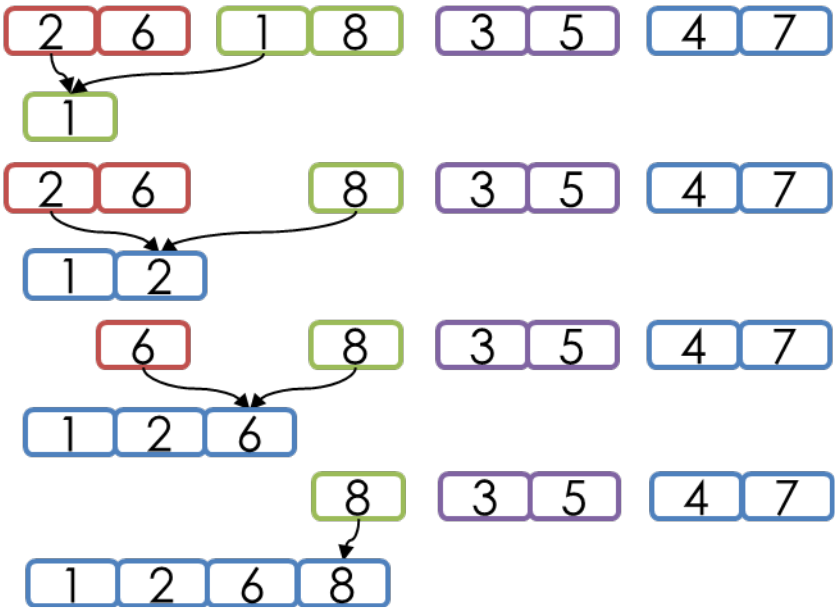


Each pair of individual items are compared and sorted into sorted pairs.



...each pair of sorted pairs are compared and sorted into sorted sets of 4.

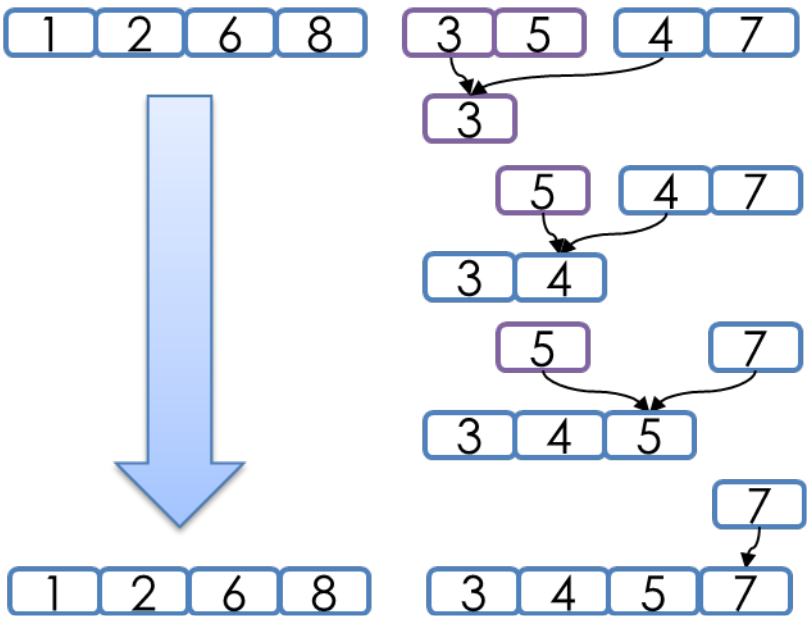
This is done by continually comparing the left most items in each pair and adding the smallest of these items to a new list each time.



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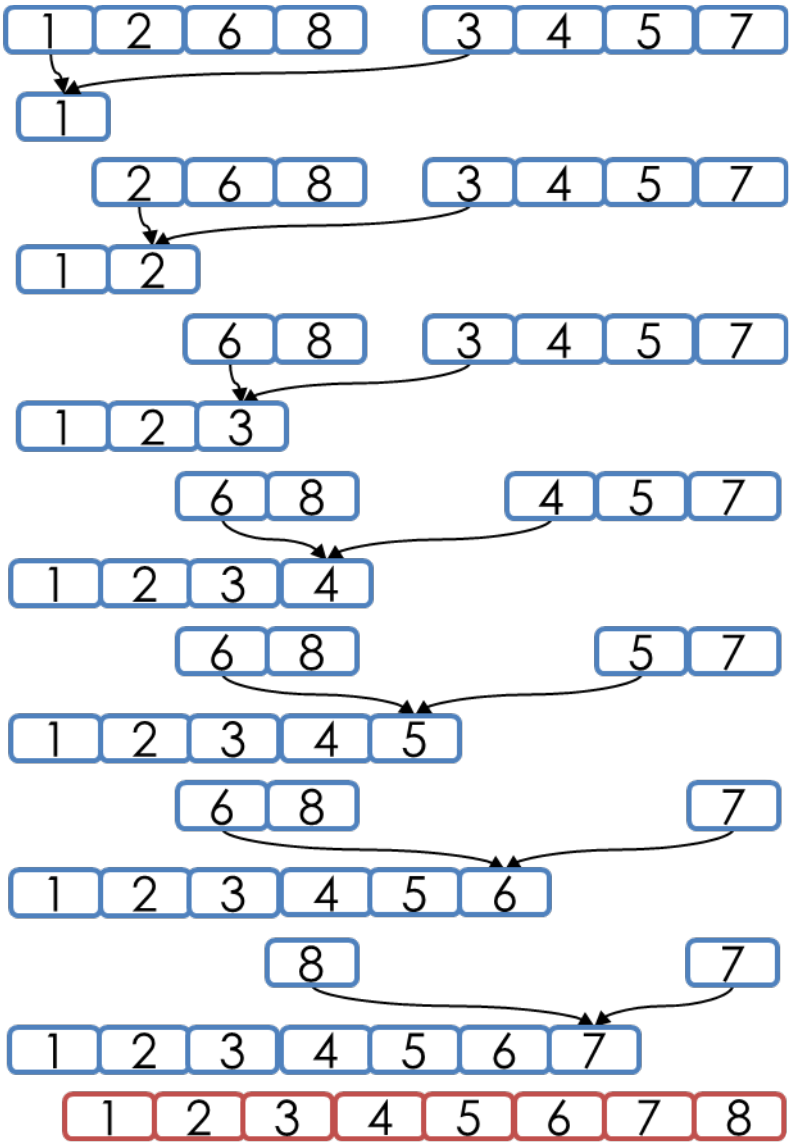
Continuing on from the previous page…
…each pair of sorted pairs are compared and sorted into sorted sets of 4.

This is done by continually comparing the left most items in each pair and adding the smallest of these items to a new list each time.



Next, the items in each pair of sorted 4s are compared and sorted into sorted sets of 8.

This is done by continually comparing the left most items in each pair and adding the smallest of these items to a new list each time.



Because this example contained 8 numbers, the data is sorted at the end of this phase. For longer data sets, the process would continue in the same manner.

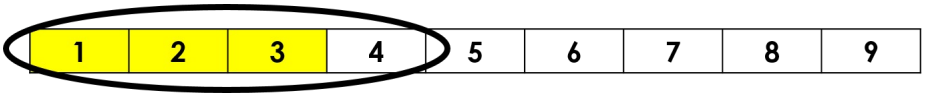
Merge Sort Algorithm Logic:

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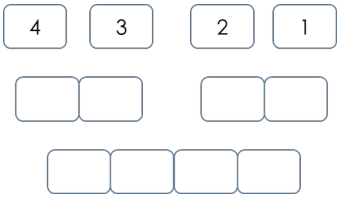
1. IF the list has only one item, it is already sorted, return.
2. Repeatedly divide the list into halves, until it can be divided no more.
3. Repeatedly join the smaller lists into new list, in sorted form.

Questions *(The question zone you choose must either match your target grade or be higher!)*

Question Zone 1-3



1. What is the purpose of the 'Merge Sort Algorithm'?. [1]
2. Copy and complete the following diagram showing (with arrows) how the a merge sort would complete the ordering of the numbers: [4, 3, 2, 1]. [3]

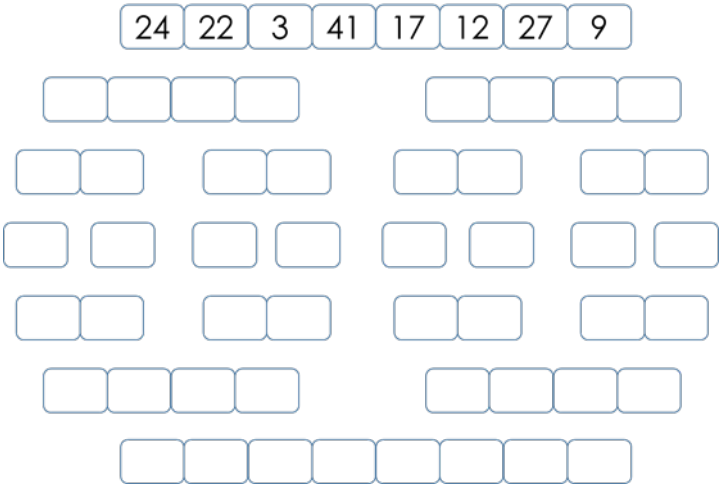


3. Describe how the 'Merge Sort' works. [3]

Question Zone 4-6

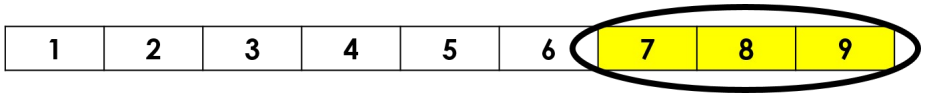


1. Describe how the 'Merge Sort' works. [3]
2. Copy and complete the following diagram showing (with arrows) how the merge sort will order the following dataset: [24, 22, 3, 41, 17, 12, 27, 9]. [3]



3. Demonstrate how a merge sort will order the following dataset: [9,8,7,6,5,4,3,2].[5]

Questions Zone 7-9



1. Demonstrate how a merge sort will order the following dataset: [9,8,7,6,5,4,3,2].[5]
2. Demonstrate how a merge sort will order the following dataset: [19,21,7,22,15,8].[5]
3. Out of the Bubble and Merge, which sorting algorithm is more efficient? Explain your answer fully [5]

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Checklist:

- Date and title, clearly presented
- Spelling & grammar checked
- Question numbers in the margin
- Handwriting neat & legible
- Punctuation / Capital letters

Keywords / Key Terms:

Merge Sort: *A method of sorting a set of data by splitting the dataset up into its individual items and then continually pairing items from the dataset in order until the entire dataset has been ordered.*

Merge: *The processes of combining elements into a single entity.*

Algorithm*: A set of steps/instructions needed to complete a software task.*

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Stick answer sheet here

State/Identify/Give/Name: Simply label a diagram, fill out a table or write a few words
Describe: Describing is 'saying what you see' (E.G.: A computer will have a CPU, Primary and Secondary storage etc)
Explain: Explaining is 'saying WHY/HOW something is like that'. (E.G.: A computer will have a CPU so that it can process all of the data the computer needs to perform a range of tasks. Primary and Secondary storage is needed because...)
Discuss: Discussing is 'looking at two sides of an issue, weighing up the two views and giving a conclusion'. Often these require a mini essay answer. (E.G.: New technology could be seen as being bad for the environment because..., but on the other hand, new technology has led to... In conclusion I believe that...)
Describe/Explain/Discuss **using examples**: Finally, if you are asked to give examples in any of these types of questions – YOU MUST **GIVE EXAMPLES**!

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Reflections **Score:** / **Percentage:** % **Grade:** **Progress** On / Above / Below

What Went

- I demonstrated a good level of understanding.
- I responded to the command words effectively.
- My answers were detailed / were written in depth.
- My work was well presented / legible.
- My answers effectively incorporated technical terminology.
- My responses were well structured / organised.
- My revision strategy was effective as I showed depth of understanding in my answers.
- My answers contained enough points / examples / explanations to achieve the marks available.

Even Better

- My answers need to be more accurate.
- I must respond correctly to the command words.
- My answers need more detail / greater depth.
- I must take greater care over my work / write neatly.
- I must incorporate key terminology into my answers.
- I must better organise my answers to improve its clarity.
- I need to improve my revision strategy, as I did not demonstrate a depth of understanding in my answers.
- My answers didn't contain enough points / examples / explanations to achieve the marks available.

Further thoughts: